We rely on unrelated third parties to host on their computers our EFT Arcus product. We have contracts and/or service agreements in place with those third parties that we believe provide us the ability to continue delivering those products without interruption.

Research and Development

To keep pace with client and market demand, we maintain an ongoing program of new product development.

Our software engineers are responsible for creating and building our software products. They do so by combining their expertise with input from our sales, marketing and product management groups as to market trends and needs. Our software engineers design and write software and manage its testing and quality assurance. We utilize third-party software developers both domestically and overseas working under our supervision to supplement our software engineers. Using these external software developers in a strategic manner allows us to access highly skilled labor pools, maintain a 24-hour development schedule, decrease time to market, and minimize programming costs.

All phases of research and development, or R&D, including scope approval, functional and implementation design, object modeling and programming, are subject to extensive internal quality assurance testing. We maintain an ongoing focus on improving our quality assurance testing infrastructure and practices. Technical reporting and client support feedback confirm the continuing positive effect of our ongoing enhancement of research and development and quality assurance processes.

Our EFT Arcus product is hosted by third-party cloud services providers. We rely upon those third parties, such as Microsoft Azure, for the continued development and enhancement of their cloud services infrastructures on which our products are hosted. We do not perform significant research and development of cloud services infrastructures using our own personnel.

Our R&D expenditures profile has been as follows ($ in thousands):

	Year ending December 31,	
	2019	2018
R&D expensed	$ 1,355	$ 1,883
Capitalized software development costs	1,074	1,276
Total resources expensed for R&D	$ 2,430	$ 3,159

Our total R&D expenditures decreased 23% in 2019 as compared to 2018 primarily due to fewer employed software engineers and technical personnel.

Total resources expended for R&D serves to illustrate our total corporate efforts to improve our existing products and to develop new products regardless of whether or not our expenditures for those efforts were expensed or capitalized. Total resources expended for R&D is not a measure of financial performance under GAAP and should not be considered a substitute for R&D expense and capitalized software development costs individually. While we believe the non-GAAP total resources expended for R&D amount provides useful supplemental information regarding our overall corporate product improvement and new product creation activities, there are limitations associated with the use of this non-GAAP measurement. Total resources expended for R&D is a non-GAAP measure not prepared in accordance with GAAP and may not be comparable to similarly titled measures of other companies since there is no standard for preparing this non-GAAP measure. As a result, this non-GAAP measure of total resources expended for R&D has limitations and should not be considered in isolation from, or as a substitute for, R&D expense and capitalized software development costs individually.

Competition

The managed file transfer software market sector is highly competitive, subject to rapid change, and significantly affected by new product introductions and other activities of market participants.

Some of our competitors in certain markets have greater financial, technical, sales, marketing and other resources than we do. Because of these and other factors, competitive conditions in these industries are likely to continue to intensify in the future. Increased competition could result in price reductions, reduced net revenue and profit margins and loss of market share, any of which could harm our business. See “Risk Factors – Risks Related to Our Operations” for further discussion of risks regarding competition.